

Linear Regression

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Suppose we are given a dataset of n observations

$$\mathbf{X} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix}$$

with the corresponding target values

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}.$$

Linear regression uses a linear function

$$f(\mathbf{x}) = \mathbf{w}^\top \mathbf{x} + b,$$

where $\mathbf{x}$ is the feature vector, $\mathbf{w}$ is the coefficient vector, and b is the intercept.

For all observations, the predicted values can be written as

$$\hat{\mathbf{y}} = \mathbf{X}\mathbf{w} + b\mathbf{1},$$

where $\hat{\mathbf{y}}$ is the vector of predictions and $\mathbf{1}$ is an n -dimensional vector of ones.

To simplify the notation, we incorporate the intercept term into the design matrix by appending a column of ones to $\mathbf{X}$. The augmented design matrix is

$$\tilde{\mathbf{X}} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} & 1 \\ x_{21} & x_{22} & \cdots & x_{2p} & 1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} & 1 \end{bmatrix}.$$

Similarly, we augment the coefficient vector as

$$\tilde{\mathbf{w}} = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_p \\ b \end{bmatrix}.$$

For simplicity, we will continue to denote the augmented design matrix and coefficient vector by $\mathbf{X}$ and $\mathbf{w}$, respectively. Then the prediction function can then be written as

$$\hat{\mathbf{y}} = \mathbf{X}\mathbf{w}.$$

The aim of linear regression is to minimize the following loss function

$$L(\mathbf{w}) = \|\mathbf{y} - \mathbf{X}\mathbf{w}\|^2.$$

Equivalently, we seek the parameter vector that minimizes this loss:

$$\hat{\mathbf{w}} = \underset{\mathbf{w}}{\operatorname{argmin}} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|^2.$$

Expanding the objective function gives

$$\|\mathbf{y} - \mathbf{X}\mathbf{w}\|^2 = (\mathbf{y} - \mathbf{X}\mathbf{w})^\top (\mathbf{y} - \mathbf{X}\mathbf{w}).$$

Taking the derivative with respect to $\mathbf{w}$ and setting it equal to zero yields

$$\mathbf{X}^\top \mathbf{X}\mathbf{w} = \mathbf{X}^\top \mathbf{y}.$$

These are known as the normal equations. Assuming $\mathbf{X}^\top \mathbf{X}$ is invertible, the solution is

$$\hat{\mathbf{w}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}.$$